

**University of Toronto at Scarborough
Department of Computer & Mathematical Sciences**

NOTES:

- There are 9 numbered pages the exam. It is your responsibility to ensure that, at the start of the exam, this booklet has all its pages. The last page is intentionally left blank.
- Answer all questions. Explain and justify your answers.
- **Show all your work.** Credit will not be given for numerical answers if the work is not shown. If you need more space use the last page of the test which is intentionally left blank.
- Do NOT continue your solutions on the back of test pages (these will not be graded).
- No cell phones and any type of e-mail or instant messaging devices are allowed to be brought to the exam. Be sure that if you have any, that they are OFF and in your backpack away from you.
- No scrap paper or any sort of paper may be brought to your exam seat. All necessary space needed for your solutions is included in the exam. The backs of test papers may be used as scrap paper.
- The use of a calculator is **NOT** permitted.

SIGNATURE:_____

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED TO DO SO.

1. **[12 marks] TRUE OR FALSE** Carefully read each statement. If the assertion must be true, then **circle T** (for true). Otherwise, **circle F** (for false). Justification is neither required nor rewarded, but a small workplace is given for your rough work. Each correct answer earns 2 points and each incorrect or blank answer earns 0 points.

- (a) Suppose that f is continuous everywhere. Let $n \in \mathbb{N}$, $a, b \in \mathbb{R}$, $a < b$. Let L_n , R_n and M_n denote the left, right and midpoint Riemann sums, respectively, for f over $[a, b]$. Then

$$\int_a^b f(x)dx = \lim_{n \rightarrow \infty} L_n = \lim_{n \rightarrow \infty} R_n = \lim_{n \rightarrow \infty} M_n.$$

T or F

- (b) Consider the integral $I = \int_1^2 e^{-x^2} dx$. Then every right Riemann sum approximation for I must be a strict underestimate.

T or F

- (c) Let $a, b \in \mathbb{R}$, $a < b$. Suppose that f is a bounded function on $[a, b]$. If there is *some* partition P of $[a, b]$ such that $U(f, P) = -4$ and $L(f, P) = -5$ then f must be integrable on $[a, b]$.

T or F

- (d) Let $n \in \mathbb{Z}^+$. If a_k is a positive-valued function defined for $k \in \mathbb{Z}^+$, then

$$\sum_{k=1}^n a_k^2 = \sum_{k=1}^n a_k a_k = \left(\sum_{k=1}^n a_k \right)^2.$$

T or F

- (e) Let $c, d \in \mathbb{R}$, $c < d$. If f and g are real-valued functions that are not integrable on $[c, d]$ then the function $f + 2g$ is also not integrable on $[c, d]$.

T or F

- (f) Suppose that g is at least twice differentiable on $\mathbb{R}$. If $g(0) = 1$ then

$$g(4) = \int_0^4 g'(x) dx + 1.$$

T or F

2. **[10 marks]** Let $a, b, c \in \mathbb{R}$ such that $a < b$. Use ONLY the Riemann definition of the definite integral to prove : if f and g are continuous on $[a, b]$ then

$$\int_a^b (f(x) - g(x) + c) \, dx = \int_a^b f(x) \, dx - \int_a^b g(x) \, dx + c(b - a).$$

3. [7 marks] Express the following limit as a definite integral. Make sure to fully justify your answer.

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \left(97 - \frac{24i}{n} - \frac{48i^2}{n^2} \right) \frac{4}{n}$$

4. [3 marks] Prove : $\int_2^5 \frac{1}{1+x^5} dx \leq \int_2^5 \frac{1}{1+x^2} dx$

5. Evaluate the following integrals. Make sure to appropriately justify your solutions.

(a) [5 marks] $\int_0^1 \sqrt{2x - x^2} \, dx.$

(b) [8 marks] $\int \frac{\cos(x)}{\sin^2(x) - \sin(x) - 6} \, dx$

(c) [**7 marks**] $\int_1^3 \frac{1}{(x+1)\sqrt{x}} dx.$

(d) [**5 marks**] $\int \tan^{-1}\left(\frac{1}{x}\right) dx$

6. [10 marks] Let $h(x) = \int_{-e^x}^{2x^4+1} 6^t(1+t^6)^{-1} dt$. Compute $h'(x)$. Make sure to fully justify your solution. Do not evaluate the integral.

7. [9 marks] Let $c, d \in \mathbb{R}$, $c < d$. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a real valued function defined by

$$f(x) = \begin{cases} 0 & , \text{if } x \in \mathbb{Q} \\ -1 & , \text{if } x \notin \mathbb{Q} \end{cases}$$

Use the Darboux definition of the definite integral to prove that $e^{f(x)}$ is not integrable on $[c, d]$. Make sure to fully justify your work.

8. [7 marks] Let $a \in \mathbb{R}$. Suppose f and g are continuous everywhere functions. Prove :

If $f(x)$ is even and $g(x)$ is odd on $[-a, a]$ then $\int_{-a}^a f(x)g(x) dx = 0$.

Do not use any symmetry integration properties, e.g., do not use the property “ $\int_{-a}^a (\text{odd function}) dx = 0$ ” in your proof.

9. [7 marks] Let $a \in \mathbb{R}^+$. If $f(0) = g(0) = 0$ and f'' and g'' are continuous everywhere, prove that $\int_0^a f(x)g''(x) dx = f(a)g'(a) - f'(a)g(a) + \int_0^a f''(x)g(x) dx$.

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